

Portfolio Selection Assignment

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The portfolio selection assignment consists of two parts. For the first part, I was asked to construct a portfolio based on my pre-knowledge, which will serve as a benchmark. The second part, which is the main body of the assignment, asked me to revise my portfolio each course week based on the material of that week. I needed to motivate my decisions, which will show whether I was able to apply the material that was discussed in class.

PART I.

1.1 Task

For this assignment I was asked to construct a portfolio that consists of stocks from the S&P500. I received a hypothetical amount of \$100,000 for which I can buy stocks or obtain a risk-free interest rate of 1% (annual). Any stock I decided to buy can only be bought in a positive integer number. The risk-free interest rate I can think of as depositing money in a bank account such that any amount can be chosen as a deposit. I was asked to hand in the Excel sheet with the number of stocks held per stock, the amount invested at the risk-free interest rate, and all relative weights in the portfolio. I need to make sure it adds up to \$100,000; I cannot be short or leave money on the table. Any investment at the risk-free interest rate should also be explicitly written.

1.2 Introduction and Background

The way I construct my first portfolio is up to me and what I know so far about finance. As for my educational background, I graduated with a Bachelor of Business Administration, which consisted of rather few financial courses. To compensate for this shortcoming, I wrote several research papers on financial markets, using different methodologies. Thanks to these papers, I was hired to work as an equity analyst at a Hungarian investment bank – this is where I came from. During my work there, I had the opportunity to gain insight into different aspects of financial analyses, from reporting on individual public companies and recommending investment opportunities, to developing econometric and machine learning-based models to predict foreign exchange rates and stock prices.

1.3 First Portfolio Construction (*Week 1*)

Even though I have been developing more sophisticated models for portfolio/stock selection, I would stick to a relatively simple strategy for the first part of the assignment. One reason for this is that I want it to be a baseline model for the following weeks' solutions, while the other reason is the given constraints – e.g., working with only yearly prices.

My solution begins with obtaining the previous year's close prices, then calculating a one-year percentage return, without considering dividends. I omit stocks without price data, i.e., that were not listed on the stock exchange in the previous year. Stocks with non-positive returns are also filtered out and receive no allocation. Then, I calculate the weight for each stock by dividing the stock's return by the sum of all returns. Multiplying the weights by the allocated \$100,000 will give the dollar amount I want to invest in each stock. To get how many shares to buy, I divide the allocated amount by the latest available close price. Since I can buy only an integer number of shares, I round this value. Rounding has a slight chance to result in an allocation of more money than the specified amount; therefore, I am supposed to round these values down. However, I tested it, and it does not exceed this amount, so I decided to simply round it. After rounding and allocating the total amount of money, I invest the leftover cash into the risk-free asset. With this, I implemented a classic positive momentum strategy.

PART II.

2.1 Task

For the second part of this assignment I am asked to update the initial portfolio that I handed in at the first hand-in date. In the course, we discussed topics that could help me strengthen my personal investment portfolio. It is up to me to determine and motivate whether material that we discussed in class is appropriate to incorporate in my portfolio. I could argue it is the case or it is not, but I need to write a well-founded motivation for my decision to update my portfolio. I can also have the opportunity not to update my portfolio, but in this case, I need to explain why I do so based on the class material, this can also be supported by an analysis.

2.2 Tangency Portfolio using Mean-Variance Analysis (*Week 2*)

In the second part of the assignment, I decided to switch from my initial momentum-based portfolio to a mean-variance based tangency portfolio. In mean-variance portfolio theory, investors evaluate portfolios solely by their expected return μ and variance σ^2 . With quadratic utility (or normally distributed returns), these two parameters are sufficient statistics for choice. Preferences are monotone: higher μ is preferred, while higher σ is disliked. The goal is therefore to identify efficient portfolios, which yield the highest expected return for a given level of risk.

Let $\mu \in \mathbb{R}^N$ denote expected returns, $\Sigma \in \mathbb{R}^{N \times N}$ the positive-definite covariance matrix of asset returns, and $\mathbf{1}$ the vector of ones. For a target return $\hat{\mu}$, the minimum-variance portfolio solves

$$\min_w \frac{1}{2} w' \Sigma w \quad \text{s.t.} \quad w' \mu = \hat{\mu}, \quad \mathbf{1}' w = 1.$$

The closed-form solution is

$$w_{\hat{\mu}} = \frac{C\hat{\mu} - A}{D} \Sigma^{-1} \mu + \frac{B - A\hat{\mu}}{D} \Sigma^{-1} \mathbf{1},$$

with

$$A = 1'\Sigma^{-1}\mu, \quad B = \mu'\Sigma^{-1}\mu, \quad C = 1'\Sigma^{-1}1, \quad D = BC - A^2.$$

Varying $\hat{\mu}$ traces out the minimum-variance frontier. The upper branch of this curve is the efficient frontier, representing all portfolios with the best attainable return at each variance level.

Among all efficient portfolios, one stands out: the portfolio that offers the highest slope in mean-variance space – i.e., the greatest increase in expected return per additional unit of risk. This is the tangency portfolio.

Formally, it solves

$$\max_w \frac{w'\mu}{\sqrt{w'\Sigma w}} \quad \text{s.t.} \quad 1'w = 1,$$

which is equivalent to maximizing the portfolio Sharpe ratio when no risk-free asset is included. The solution is proportional to $t \propto \Sigma^{-1}\mu$. Scaling is determined by the normalization $1't = 1$. Thus,

$$t = \frac{\Sigma^{-1}\mu}{1'\Sigma^{-1}\mu}$$

This portfolio represents the unique combination of risky assets that achieves the steepest risk-return trade-off along the efficient frontier. All other efficient portfolios are linear combinations of this tangency portfolio and the global minimum-variance portfolio. Including a risk-free asset would alter the efficient frontier, making it a straight line from the risk-free rate to the tangency portfolio. However, in this assignment, I handle the risky assets first, and then allocate any leftover cash to the risk-free asset.

In constructing the tangency portfolio, the unconstrained mean-variance solution gave weights that included negative values (short positions) and values greater than one (leveraged longs). Since I do not want to allow short selling in this assignment, I decided on a practical modification: I set all negative weights to zero and then renormalized the positive weights so that they sum to one. This way, my portfolio remains long-only while preserving as much of the original tangency solution as possible. Although this adjustment is not strictly mean-variance optimal under the no-shorting constraint, it is a reasonable compromise that avoids excessive complexity in the optimization while ensuring realistic, fully invested allocations.

As for the allocation, I used the same approach as in the first part of the assignment, only now with the new tangency weights.

2.3 Fama-French 3-Factor Model (*Week 3*)

In the third part of the assignment, I apply the Fama-French 3-Factor model to guide portfolio construction. The model extends the traditional CAPM by including two additional sources of systematic risk: SMB (small minus big), which captures the size premium, and HML (high minus low), which captures the value premium. Formally, the excess return of asset i is modeled as:

$$R_i - R_f = \alpha_i + \beta_{iM}(R_M - R_f) + \beta_{iS}SMB + \beta_{iH}HML + \varepsilon_i,$$

where R_i is the asset return, R_f the risk-free rate, R_M the market return, and $\beta_{iM}, \beta_{iS}, \beta_{iH}$ represent the asset's exposures to the three factors.

Rather than forecasting individual stock returns, I use a factor-tilt (long-only) allocation. The idea is to set target exposures to the underlying risk factors and then choose stock weights so that the resulting portfolio achieves these exposures while remaining fully invested and long-only.

For this project I decide on the following target exposures with motivations:

- Market (β_M) = 1 $\rightarrow$ I want the portfolio to track the overall market risk, ensuring it moves broadly with market movements.
- SMB (β_S) = 0 $\rightarrow$ I do not want a systematic tilt toward small- or large-cap stocks, as size effects have been less consistent in recent years.
- HML (β_H) = 0.3 $\rightarrow$ I introduce a modest value tilt, reflecting the empirical evidence that value stocks tend to outperform growth stocks over the long run.

These choices are consistent with empirical asset pricing literature, which documents the persistence of the value premium, while also recognizing that size effects are less stable in recent decades.

To translate targets into weights, suppose B is the $N \times 3$ matrix of estimated factor loadings (betas) for the N available stocks. My portfolio weights w (an N -vector) must satisfy:

$$B^\top w \approx \beta^{\text{target}}, \quad \sum_{i=1}^N w_i = 1, \quad w_i \geq 0,$$

where

$$\beta^{\text{target}} = \begin{bmatrix} 1.0 \\ 0.0 \\ 0.3 \end{bmatrix}.$$

This means the weighted average factor exposures of the portfolio match my chosen targets, while ensuring the portfolio is fully invested and long-only.

This is a constrained optimization problem. I solve for the weights w using `cvxpy`, a convex optimization library in Python. The problem is set up as a quadratic program:

$$\min_w \|B^\top w - \beta^{\text{target}}\|_2^2$$

subject to the budget constraint $\sum_i w_i = 1$ and non-negativity constraints $w_i \geq 0$.

This optimization process finds the feasible set of stock weights that minimizes the squared distance between the achieved portfolio exposures and my desired factor exposures. If exact matching is not possible due to the long-only constraint, the optimizer finds the closest possible solution.

The resulting weights are then allocated to the \$100,000 investment, as in previous weeks, with any leftover cash invested at the risk-free rate.

2.4 Technical Indicators (*Week 4*)

In the fourth week, I apply technical indicators to guide portfolio weight allocation using monthly data. There are 5 main indicators we discussed in class, even though there are countless more. These are: Moving Averages (MA), Support and Resistance (RS), Alexander Filter (ALX), Bollinger Bands (BB), Directional Indicator (DI). I decided to go with Bollinger Bands and Directional Indicator, as this way I can also include the Moving Average, which is the basis of BB. Support and Resistance are not really applicable for portfolio selection, as they are more of a trading tool. So, my objective is to combine a measure of trend strength (DI) with a volatility-adjusted breakout filter (BB), and then construct a risk-balanced portfolio.

First, I use the Directional Indicator (DI) to quantify the strength of a stock's trend relative to its noise. With a one-year lookback ($n = 12$), the indicator is defined as:

$$DI_t = \frac{S_t - S_{t-12}}{\sum_{i=1}^{11} |S_{t-i} - S_{t-i-1}|}$$

where (S_t) is the stock price at month (t) . The numerator captures the net price change over the past year, while the denominator sums the absolute monthly price changes, representing total volatility.

A higher DI indicates a more consistent directional move rather than sideways volatility.

Second, I compute Bollinger Bands (BB), which provide a volatility-adjusted signal of whether prices are breaking out of their recent range. Using a 12-month moving average, I define:

$$Upper_t = MA_{12,t} + B \cdot \sigma_{12,t}, \quad Lower_t = MA_{12,t} - B \cdot \sigma_{12,t}$$

where $(\sigma_{12,t})$ is the standard deviation of the past 12 months of returns, and B is set to 1. The B parameter controls the width of the bands, usually set between 1 and 2. In this case, I choose $B = 1$ to be less strict and thus include more stocks in the selection.

For portfolio selection, I require two conditions:

1. The stock's DI lies in the top 50% percentile of all assets, ensuring I focus on those with relatively strong trends.
2. The stock's price is above the upper Bollinger Band, confirming a volatility-adjusted break-out.

Finally, I allocate weights to the selected assets using inverse volatility weighting. To do this, I compute annualized volatility from monthly returns:

$$\sigma_i = \sqrt{12} \cdot \text{std}(r_{i,1..12})$$

and assign weights as:

$$w_i = \frac{\frac{1}{\sigma_i}}{\sum_{j \in \mathcal{P}} \frac{1}{\sigma_j}}$$

where $(\mathcal{P})$ denotes the set of selected assets. This ensures that more stable stocks receive larger weights, while highly volatile stocks are down-weighted. The resulting weights are then allocated, as in previous weeks.

2.5 Tangency Portfolio of the ESG-efficient frontier (*Week 5*)

In the fifth part of the assignment, I extend the classical mean-variance portfolio optimization framework discussed in the second section of the tangency portfolio section by incorporating environmental, social, and governance (ESG) considerations, following Pedersen, Fitzgibbons, and Pomorski (2021). Their approach builds directly on Markowitz’s theory but adds investor preferences for ESG exposure as an additional dimension in the optimization. The key idea is that responsible investing can be achieved without abandoning the mean-variance paradigm – instead, ESG preferences are integrated into the utility function itself, creating what the authors call the ESG-efficient frontier.

This ESG-efficient frontier generalizes the standard risk-return trade-off by adding a third dimension: ESG performance. In other words, each optimal portfolio now reflects not only an expected return and a level of risk but also an ESG score. Investors can therefore choose allocations that align with their sustainability preferences while remaining on an efficiency frontier consistent with modern portfolio theory.

To formalize this, I start from the standard mean-variance optimization problem:

$$\max_w \mu^\top w - \frac{\gamma}{2} w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1,$$

where μ is the vector of expected excess returns, Σ is the covariance matrix of asset returns, and γ is the investor’s risk-aversion parameter.

Each asset is also assigned an ESG score (s_i), collected into a vector (s). Since now ESG scores are reported on a 0-100 scale, I have to handle them appropriately, e.g., by normalizing them. The most common approach in academic and applied finance when integrating non-return measures into optimization is to use z-score normalization:

$$s_i^* = \frac{s_i - \bar{s}}{\sigma_s}$$

The normalized scores vector are denoted by s^* . I then introduce a taste parameter (ϕ), which captures the strength and direction of the investor’s ESG preference. The modified optimization becomes:

$$\max_w \mu^\top w + \phi s^{*\top} w - \frac{\gamma}{2} w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1.$$

Here, $\phi > 0$ indicates a tilt toward high-ESG assets, $\phi < 0$ a tilt toward low-ESG (“sin”) assets, and $\phi = 0$ corresponds to the traditional mean-variance setup. By varying ϕ , I can trace out the ESG-efficient frontier, which shows the attainable combinations of expected return, volatility, and ESG exposure.

Analogous to the classical setting, I can also identify a single ESG-adjusted tangency portfolio – the point on the ESG-efficient frontier that maximizes the ESG-adjusted Sharpe ratio. This is obtained by treating the ESG preference as a shift in expected returns, defining:

$$\mu_{\text{ESG}} = \mu + \phi s^*.$$

The optimal portfolio weights are then:

$$w_{\text{ESG}} = \frac{\Sigma^{-1} \mu_{\text{ESG}}}{\mathbf{1}^\top \Sigma^{-1} \mu_{\text{ESG}}}.$$

This portfolio represents the optimal trade-off between expected return, risk, and sustainability preference. ϕ can be interpreted as how much return the investor is willing to give up per 1 standard deviation improvement in ESG. When $\phi = 1$, the portfolio becomes ESG-tilted, meaning that it moderately shifts weights toward assets with higher ESG scores while maintaining a relatively balanced risk-return profile. When $\phi = 2$, the investor is strongly sustainability-driven, assigning substantial importance to ESG performance even at the cost of lower expected returns or higher tracking error relative to the traditional tangency portfolio.

In my implementation, I choose $\phi = 1$ to reflect a moderate ESG preference. Then, I allocate the resulting weights to the \$100,000 investment, as in previous weeks, with any leftover cash invested at the risk-free rate.

2.6 Life-Cycle Investing (*Week 6*)

In the final part of the assignment, I apply the life-cycle investment model of Cocco, Gomes, and Maenhout (2005) to determine the optimal allocation between risky and risk-free assets, and subsequently among individual stocks. The idea is that my risk exposure should vary systematically with age, risk aversion, and human capital – the present value of future labor income.

The starting point of my allocation is the Merton fraction, which expresses the optimal share of total wealth invested in risky assets under CRRA (Constant Relative Risk Aversion) preferences:

$$\alpha^* = \frac{\mu - r_f}{\gamma \sigma^2}$$

where

- μ is the expected return on the risky asset,
- r_f is the risk-free rate,
- σ^2 is the variance of risky returns, and
- γ is the coefficient of relative risk aversion.

However, total wealth consists of both financial wealth (W_t) and human capital (H_t), the latter representing the discounted value of future income. Thus, the fraction of financial (i.e. risky) wealth to allocate to risky assets must be adjusted for human capital exposure:

$$\alpha_t^{\text{risky}} = \frac{\mu - r_f}{\gamma \sigma^2} \left(1 - \frac{H_t}{W_t + H_t} \right)$$

The intuition is that younger investors (with large H_t relative to W_t) can afford to take more financial risk, as their future income acts as a stable asset buffer.

Given that I am 24 years old, I am in the early phase of my working life – meaning my human capital is large relative to my financial wealth. I use the following baseline parameters to illustrate my life-cycle position:

- Risk-free rate: $r_f = 1\%$ (given in the assignment, annualized)
- Expected return on risky assets: $\mu \approx 14\%$ (calculated from historical data, annualized)
- Volatility of risky assets: $\sigma \approx 36\%$ (calculated from historical data, annualized)
- Relative risk aversion: $\gamma = 8$ (using Holt and Laury (2002), typical for young investors)
- Human capital to total wealth ratio: $\frac{H_t}{W_t + H_t} = 0.75$ (using Cocco et al. (2005), assuming 40 years of work ahead and stable income)

Substituting these values gives approximately:

$$\alpha_t^{\text{risky}} \approx \frac{0.14 - 0.01}{8 \times 0.36^2} \times (1 - 0.75) = 0.0328$$

Hence, about 3.28% of my total wealth (\$ 100,000 in this case) should be invested in risky assets, and 96.72% in the risk-free asset.

Once the total equity share α_t^{risky} is determined, I allocate this portion across the individual stocks in my dataset. Using historical price data, I compute average returns $\hat{\mu}_i$ and the covariance matrix Σ of stock returns. I then construct a tangency portfolio among equities: $\tilde{w} = \frac{\Sigma^{-1}\hat{\mu}}{\mathbf{1}^T \Sigma^{-1} \hat{\mu}}$

Finally, I scale these relative weights by my total risky share:

$$w_i = \alpha_t^{\text{risky}} \times \tilde{w}_i, \quad w_{\text{bond}} = \alpha_t^{\text{risk-free}} = 1 - \alpha_t^{\text{risky}}$$

This yields my final portfolio composition, combining the theoretical life-cycle allocation with data-driven portfolio construction.

At age 24, my large human capital implies that my exposure to financial risk should remain moderate but not negligible. As I age and human capital declines relative to financial wealth, the term $1 - \frac{H_t}{W_t + H_t}$ increases, leading to a higher equity share earlier in life that gradually decreases toward retirement. This dynamic allocation captures the essence of life-cycle investing: balancing future income stability with the need for long-term capital growth.

CONCLUSION

Throughout this assignment, my portfolio construction evolved from a simple rule-based strategy into a coherent sequence of theory-driven frameworks. Each step reflected my growing understanding of how different quantitative concepts translate into practical portfolio decisions.

I began with a basic momentum strategy, which provided a transparent benchmark based purely on past returns. Moving to the mean-variance tangency portfolio helped me internalize the risk-return trade-off and the logic of efficient frontiers, while adapting the theoretical model to a realistic long-only setting. The Fama-French factor-based approach then shifted my focus from individual securities to systematic risk exposures, illustrating how portfolio returns can be managed through factor tilts. The technical indicator method combined trend-following and volatility-adjusted breakouts, allowing me to link behavioral price patterns with quantitative risk balancing. Incorporating ESG preferences extended the mean-variance framework into three dimensions – return, risk, and sustainability – showing how investor values can be integrated without losing analytical rigor. Finally, the life-cycle model introduced a dynamic, intertemporal perspective: optimal risk exposure should evolve with age, risk aversion, and human capital, tying portfolio choice directly to personal circumstances.

Looking back, these approaches form a natural progression – from empirical intuition to structured optimization and, ultimately, to intertemporal asset allocation. If I were to apply one framework in practice, I would combine the mean-variance foundation with life-cycle principles to ensure risk evolves consistently with changing wealth and human capital. More broadly, this assignment taught me that quantitative investing is not about finding a single optimal model, but about understanding the trade-offs each framework implies and applying them coherently to real-world decisions.

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